3.1 Integration by parts

$\int u\,dv = uv - \int v\,du$

$\int \ln x \cdot \frac{1}{x}\,dx$ $\quad u = \ln x$, $du = \frac{1}{x}\,dx$

$= \int u\,du$

$= \frac{u^2}{2} + C = \frac{\ln x^2}{2} + C$

$\int x \ln x\,dx$ $\quad u = \ln x$, $du = \frac{1}{x}\,dx$, $v = \frac{x^2}{2}$, $dv = x$

$= \ln x \frac{x^2}{2} - \int \frac{x^2}{2}\frac{1}{x}\,dx$

$= \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\frac{1}{x}\,dx$

$= \frac{x^2}{2}\ln x - \frac{1}{2}\int x\,dx$

$= \frac{x^2}{2}\ln x - \frac{1}{2}\frac{x^2}{2} + C$

$= \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$

$\int x\sin(x)\,dx$ $\quad u = x$, $du = 1\,dx$, $v = -\cos x$, $dv = \sin x\,dx$

$= x - \cos x - \int -\cos x\,dx$

$= -x\cos x + \sin x + C$

$\int x\tan^2(x)\,dx$ $\quad u = x$, $du = dx$, $v = \sec^2 x - 1$, $v = \tan x - x$

$= x\tan x - x^2 - \int \tan x - x\,dx$

$= x\tan x - x^2 - \int \frac{\sin x}{\cos x}dx + \int x\,dx$

$\frac{\sin x}{\cos x}dx$

$\int \frac{-du}{u}$ $\quad u = \cos x$, $du = -\sin x$, $-du = \sin x$

$= -\int \frac{1}{u}\,du$

$= -\ln u = -\ln \cos x$

$= x\tan x - x^2 + \ln\cos x + \frac{x^2}{x} + C$ $\quad dv = \tan^2 x\,dx$

$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$

$\tan^2 x + 1 = \sec^2 x$

$\tan^2 x = \sec^2 x - 1$

$\int x e^x\,dx$ $\quad u = x$, $du = 1\,dx$, $v = e^x$, $dv = e^x\,dx$

$= xe^x - \int e^x\,dx$

$= xe^x - e^x + C$

$\int x^2 e^x\,dx$ $\quad u = x^2$, $du = 2x$, $v = e$, $dv = e\,dx$

$= e^x x^2 - \int e^x\,2x\,dx$

$= e^x x^2 - 2\int e^x x\,dx$

$= e^x x^2 - 2(xe^x - e^x) + C$

$\int \ln(x)\,dx$ $\quad u = \ln x$, $du = \frac{1}{x}$, $v = x$, $dv = dx$

$= \ln x\, x - \int x\frac{1}{x}\,dx$

$= x\ln x - \int dx$

$= x\ln x - x + C$

$\int \cos(x)\,e^x\,dx$ $\quad u = e^x$, $du = e^x\,dx$, $v = \sin x$, $dv = \cos x\,dx$

$= e^x \sin x - \int \sin x\, e^x\,dx$

$\int \sin x\, e^x\,dx$ $\quad u = e^x$, $du = e^x\,dx$, $v = -\cos x$, $dv = \sin x$

$= -e^x\cos x + \underbrace{\int \cos x\, e^x\,dx}_{\text{Original Q}}$

$= e^x \sin x - [-e^x\cos x + \int \cos e^x\,dx]$

$\int \cos(x)e^x = e^x\sin x + e^x\cos x - \int \cos x\, e^x\,dx$

$2\int \cos(x)e^x\,dx = e^x\sin x + e^x\cos x$

$\int \cos(x)e^x\,dx = \frac{e^x\sin x + e^x\cos x}{2} + C$

$\int x^3 e^{x^2}\,dx$ $\quad t = x^2$, $dt = 2x$, $\frac{dt}{2} = x\,dx$

$= \int x^3 e^{x^2}\,dx$

$= \int x^2 e^{x^2} x\,dx$

$= \int t e^t \frac{dt}{2}\,dx$

$= \frac{1}{2}\int t e^t\,dt$

$= \frac{1}{2}(te^t - e^t) + C$

$= \frac{1}{2}(x^2e^{x^2} - e^{x^2}) + C$

$\int \tan^{-1}(x)\,dx$ $\quad u = \tan^{-1}x$, $du = \frac{1}{1+x^2}dx$, $v = x$, $dv = dx$

$= \tan^{-1}x\, x - \int x\frac{1}{1+x^2}\,dx$

$= x\tan^{-1}x - \int \frac{x}{1+x^2}\,dx$

$= x\tan^{-1}x - \frac{1}{2}\int \frac{2x}{1+x^2}\,dx$

$\int \frac{2x}{1+x^2}$ $\quad u = 1+x^2$, $du = 2x$

$\int \frac{du}{u} = \ln|1+x^2|$

$= x\tan^{-1}x - \frac{1}{2}\ln|1+x^2| + C$

Integration by Parts of definite Integral

$$\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$$

$$\int_1^e \frac{\ln(x)}{x^2}\,dx$$

$u = \ln x$

$du = \frac{1}{x}$

$v = \int x^{-2}dx = \frac{x^{-1}}{-1} = -\frac{1}{x}$

$dv = \frac{1}{x^2}$

$$= \ln x - \frac{1}{x} - \int -\frac{1}{x}\,\frac{1}{x}$$

$$= -\frac{\ln x}{x} + \int \frac{1}{x^2}$$

$$= \frac{-\ln x}{x} - \frac{1}{x}$$

$$\therefore \int_1^e \frac{\ln(x)}{x^2}\,dx = -\frac{\ln x}{x} - \frac{1}{x}\Big|_1^e$$

$$= -\frac{\ln e}{e} - \frac{1}{e} - \left(\frac{\cancel{\ln 1}\ 0}{1} - \frac{1}{1}\right)$$

$$= -\frac{1}{e} - \frac{1}{e} - (-1)$$

$$= -\frac{2}{e} + 1$$

$$\int_1^4 t\ln\sqrt{t}\,dt$$

$z = \sqrt{t} \rightarrow z^2 = t$

$2z\,dz = dt$

$$\int t\ln\sqrt{t}\,dt = \int z^2 \ln z\ 2z\,dz$$

$$= \int 2z^3 \ln z\,dz$$

let $u = \ln z$ $\quad v = \frac{2z^4}{4} = \frac{z^4}{2}$

$du = \frac{1}{z}$ $\quad dv = 2z^3 dz$

$$\ln z\,\frac{z^4}{2} - \int \frac{z^4}{2}\,\frac{1}{z}\,dz$$

$$\ln z\,\frac{z^4}{2} - \int \frac{z^3}{2}\,dz$$

$$= \frac{z^4}{2}\ln z - \frac{z^4}{8}\,dz$$

$$= \frac{(z^2)^2}{2}\ln z - \frac{(z^2)^2}{8}\,dz$$

$$= \frac{t^2}{2}\ln\sqrt{t} - \frac{t^2}{8}$$

$$= \frac{(4)^2}{2}\ln\sqrt{4} - \frac{4^2}{8} - \left(\frac{(1)^2}{2}\ln\sqrt{1} - \frac{(1)^2}{8}\right)$$

$$= 8\ln\sqrt{4} - 2 - \left(\cancel{\frac{1}{2}\ln 1}_{\,0} - \frac{1}{8}\right) = 8\ln 2 - 2 + \frac{1}{8} = 8\ln 2 - \frac{15}{8}$$